

Start coding or [generate](#) with AI.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df = pd.read_csv('superstore.csv', encoding='cp1252')
df.head()
```

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	...	Postal Code	Region	Product ID
0	1	CA-2016-152156	11-08-2016	11-11-2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	...	42420	South	FUR-BO-10001798
1	2	CA-2016-152156	11-08-2016	11-11-2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	...	42420	South	FUR-CH-10000454
2	3	CA-2016-138688	06-12-2016	6/16/2016	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	...	90036	West	OFF-LA-10000240
3	4	US-2015-108966	10-11-2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	...	33311	South	FUR-TA-10000577
4	5	US-2015-108966	10-11-2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	...	33311	South	OFF-ST-10000760

5 rows × 21 columns

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9994 entries, 0 to 9993
Data columns (total 21 columns):
#   Column              Non-Null Count  Dtype
---  -
0   Row ID              9994 non-null  int64
1   Order ID           9994 non-null  object
2   Order Date         9994 non-null  object
3   Ship Date          9994 non-null  object
4   Ship Mode          9994 non-null  object
5   Customer ID        9994 non-null  object
6   Customer Name      9994 non-null  object
7   Segment            9994 non-null  object
8   Country            9994 non-null  object
9   City               9994 non-null  object
10  State              9994 non-null  object
11  Postal Code        9994 non-null  int64
12  Region            9994 non-null  object
13  Product ID         9994 non-null  object
14  Category           9994 non-null  object
15  Sub-Category       9994 non-null  object
16  Product Name       9994 non-null  object
17  Sales              9994 non-null  float64
18  Quantity           9994 non-null  int64
19  Discount           9994 non-null  float64
20  Profit             9994 non-null  float64
dtypes: float64(3), int64(3), object(15)
memory usage: 1.6+ MB
```

```
df.describe()
```

	Row ID	Postal Code	Sales	Quantity	Discount	Profit	
count	9994.000000	9994.000000	9994.000000	9994.000000	9994.000000	9994.000000	
mean	4997.500000	55190.379428	229.858001	3.789574	0.156203	28.656896	
std	2885.163629	32063.693350	623.245101	2.225110	0.206452	234.260108	
min	1.000000	1040.000000	0.444000	1.000000	0.000000	-6599.978000	
25%	2499.250000	23223.000000	17.280000	2.000000	0.000000	1.728750	
50%	4997.500000	56430.500000	54.490000	3.000000	0.200000	8.666500	
75%	7495.750000	90008.000000	209.940000	5.000000	0.200000	29.364000	
max	9994.000000	99301.000000	22638.480000	14.000000	0.800000	8399.976000	

```
df.isnull().sum()
```

	0
Row ID	0
Order ID	0
Order Date	0
Ship Date	0
Ship Mode	0
Customer ID	0
Customer Name	0
Segment	0
Country	0
City	0
State	0
Postal Code	0
Region	0
Product ID	0
Category	0
Sub-Category	0
Product Name	0
Sales	0
Quantity	0
Discount	0
Profit	0

```
dtype: int64
```

```
df.shape
```

```
(9994, 21)
```

```
df.fillna(df.mean(numeric_only=True), inplace=True)
```

```
df.dropna(inplace=True)
```

```
print(df.duplicated().sum())
```

```
0
```

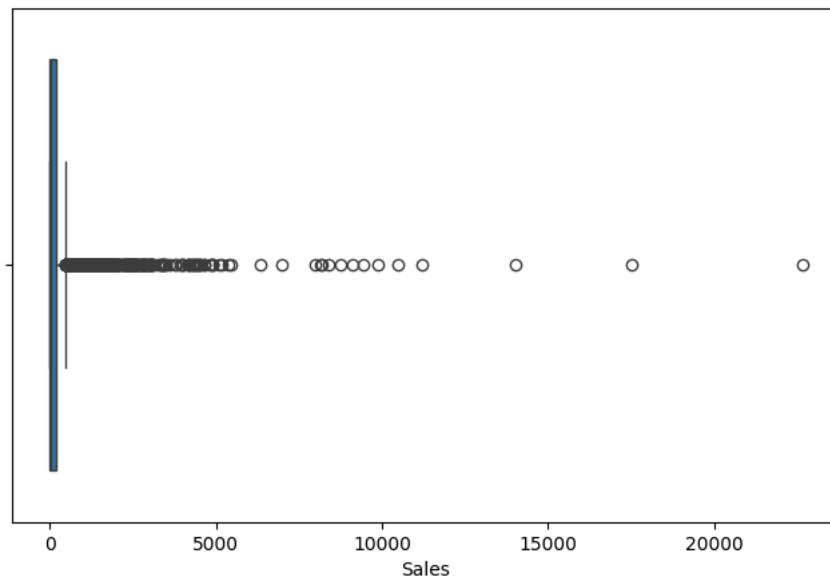
```
print(df.duplicated().sum())
```

```
0
```

```
print(df.duplicated().sum())
```

```
0
```

```
plt.figure(figsize=(8,5))
sns.boxplot(x=df['Sales'])
plt.show()
```



```
Q1 = df['Sales'].quantile(0.25)
Q3 = df['Sales'].quantile(0.75)

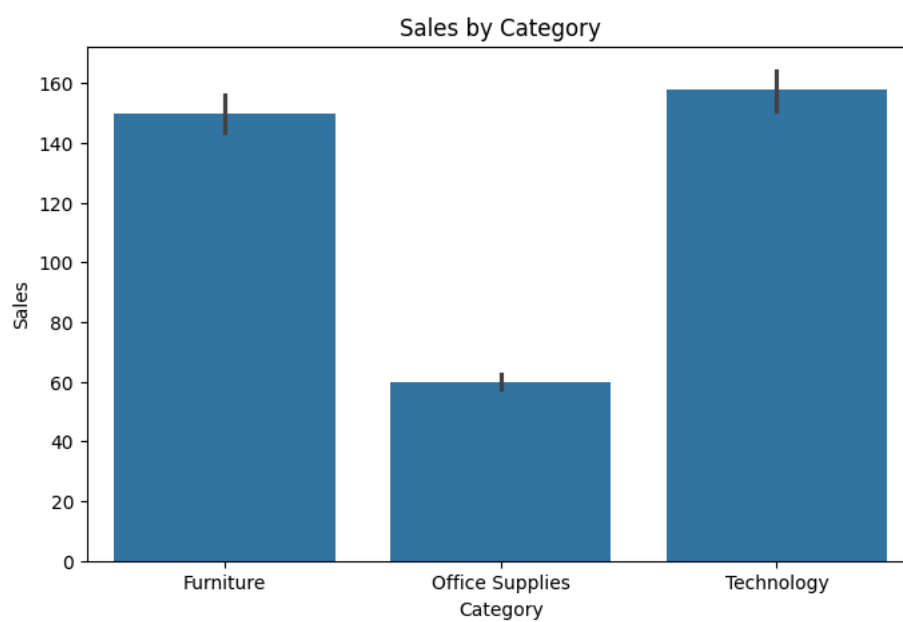
IQR = Q3 - Q1

df = df[
    (df['Sales'] >= Q1 - 1.5*IQR) &
    (df['Sales'] <= Q3 + 1.5*IQR)
]
```

```
plt.figure(figsize=(8,5))

sns.barplot(
    x='Category',
    y='Sales',
    data=df
)

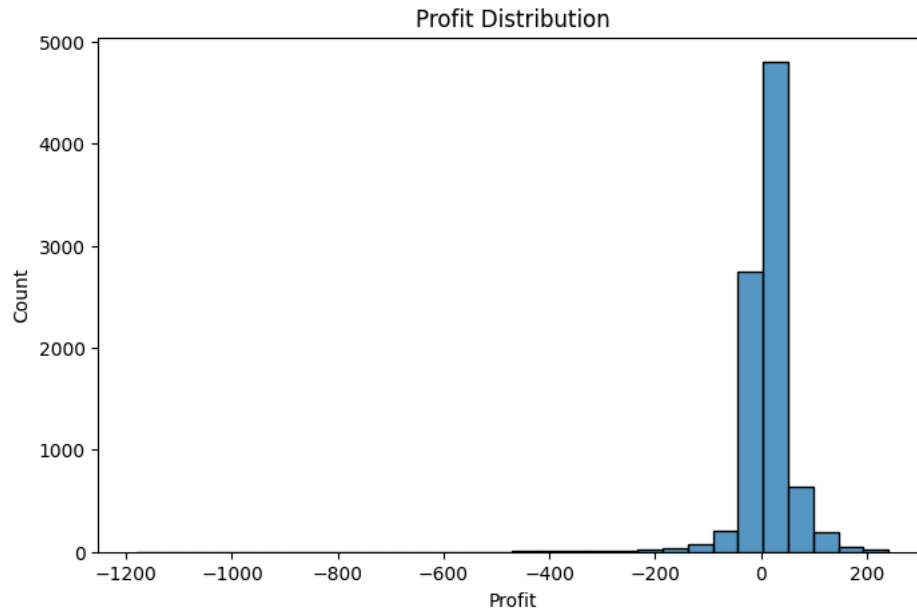
plt.title('Sales by Category')
plt.show()
```



```
plt.figure(figsize=(8,5))

sns.histplot(
    df['Profit'],
    bins=30
```

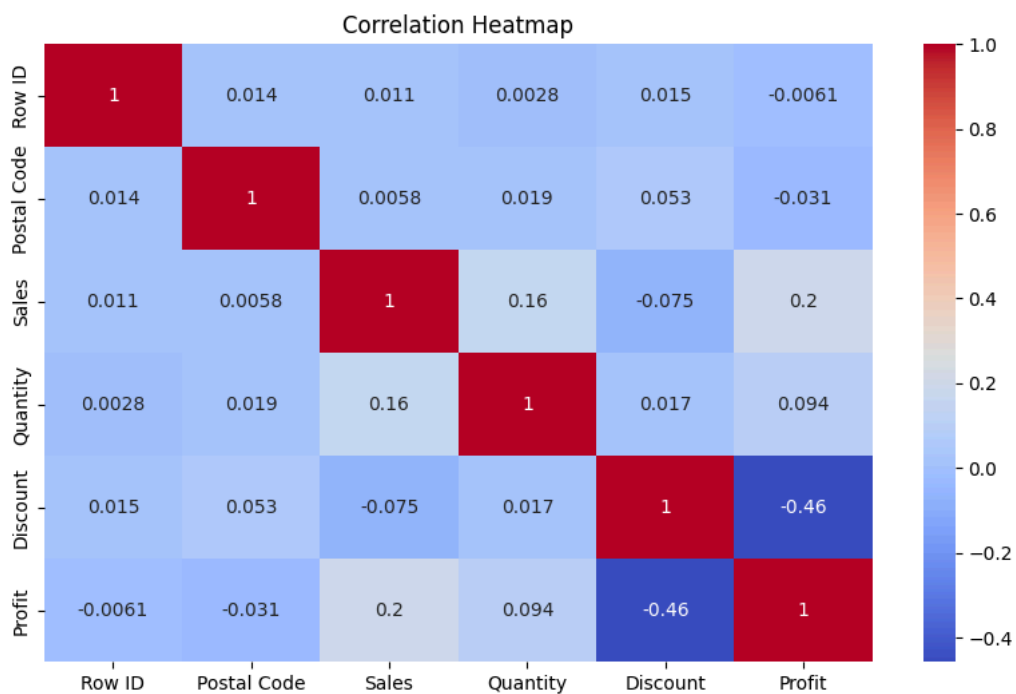
```
)
plt.title('Profit Distribution')
plt.show()
```



```
plt.figure(figsize=(10,6))

sns.heatmap(
    df.corr(numeric_only=True),
    annot=True,
    cmap='coolwarm'
)

plt.title('Correlation Heatmap')
plt.show()
```



Start coding or [generate](#) with AI.

Start coding or [generate](#) with AI.

